Summary in Graph

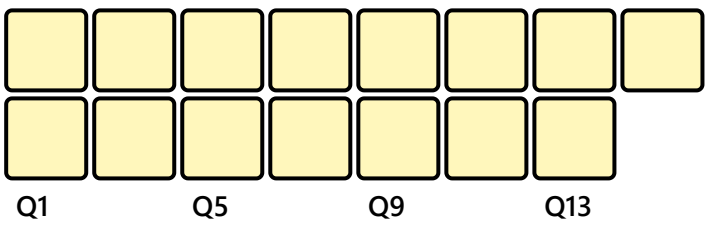
Exam Summary (GO Classes GATE CS | Theory of Computation | Finite Automata | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (55) out of 545 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Theory of Computation

Minimum number of states required in DFA accepting binary strings not ending in "101" is _____.

A. 3
B. 4
C. 5
D. 6

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Given two DFA's $\mathrm{M1}$ and $\mathrm{M2}$. They are equivalent if

A. $\mathrm{M1}$ and $\mathrm{M2}$ has the same number of states
B. $\mathrm{M1}$ and $\mathrm{M2}$ accepts the same language i.e $\mathrm{L(M1)} = \mathrm{L(M2)}$
C. $\mathrm{M1}$ and $\mathrm{M2}$ has the same number of final states
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The number of 8-bit strings beginning with either 111 or 101 is _____.

A. 64
B. 128
C. 265
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider Language A defined over the alphabet $\Sigma = \{0, 1\}$ as $\mathrm{A} = \{0^{\lfloor n/2 \rfloor} 1^n : n \geq 0\}$

The expression $\lfloor n/2 \rfloor$ means the floor of $n/2$, or what you get by rounding $n/2$ down to the nearest integer.

Which of the following is not an example of a string in A?

A. 011
B. 0111
C. 0011
D. 001111

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Assume L is a language over the alphabet Σ.

Assume λ denotes the empty string.

Which of the following statements is/are always true?

A. $\lambda \notin \mathrm{L}^+$
B. $\lambda \in \mathrm{L}^*$
C. $\lambda \notin \Sigma^+$
D. $\lambda \in \Sigma^*$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Assume L is a language over the alphabet Σ.

Assume ϵ denotes the empty string.

Which of the following statements is/are always true?

A. $\Sigma^* - \{\epsilon\} = \Sigma^+$
B. $\mathrm{L}^* - \{\epsilon\} = \mathrm{L}^+$
C. $\Sigma^* = \Sigma^+ \cup \{\epsilon\}$
D. $\mathrm{L}^* = \mathrm{L}^+ \cup \{\epsilon\}$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements is/are false?

A. If $\mathrm{L} = \{w \in \{0,1\}^* : \text{w has an unequal number of } 0\text{'s and } 1\text{'s }\}$, then $\mathrm{L}^* = \{0,1\}^*$.
B. Let $\Sigma = \{a, b\}$. There are more than 20 words of length 4 over Σ.
C. For any languages $\mathrm{L}_1, \mathrm{L}_2, (\mathrm{L}_1 \cup \mathrm{L}_2) \cap \mathrm{L}_1 = \mathrm{L}_1$.
D. For any languages $\mathrm{L}_1, \mathrm{L}_2, \mathrm{L}_1^* = \mathrm{L}_2^*$ iff $\mathrm{L}_1 = \mathrm{L}_2$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Assume Λ denotes the empty string.

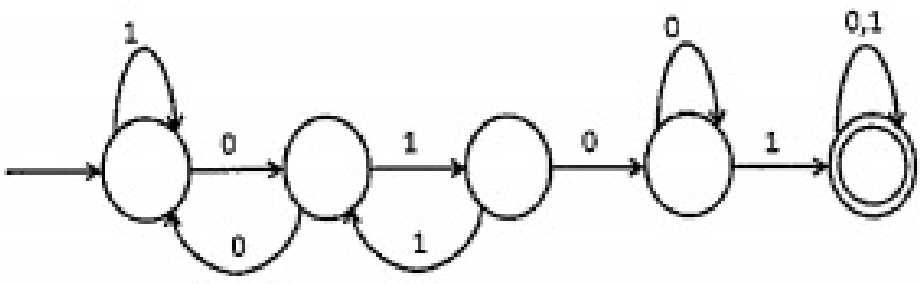
Which of the following statements is/are true?

A. For any languages $\mathrm{L}_1, \mathrm{L}_2, (\mathrm{L}_1 \cup \mathrm{L}_2)^* = \mathrm{L}_1^*$.
B. Every string of length 4 over the alphabet $\{a, b\}$ contains the substring xx, for some non-null string x.
C. For every language L, $\mathrm{LL}^* = \mathrm{L}^*$ if and only if $\Lambda \in \mathrm{L}$.
D. If $\mathrm{L}_1 \subseteq \mathrm{L}_2$, then $\mathrm{L}_1^* \subseteq \mathrm{L}_2^*$.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following Deterministic Finite Automaton M.



Let S denote the set of eight-bit strings whose second, third, sixth, and seventh bits are 1. The number of strings in S that are accepted by M is _____.

A. 0
B. 1
C. 2
D. 3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Assume Λ denotes the empty string.

Suppose $\mathrm{L} \subseteq \{a, b\}^*$ is defined as follows:

$\Lambda \in \mathrm{L}$; for every $x \in \mathrm{L}$, both xa and xba are in L.

For any string w in L, which of the following statements is/are definitely true?

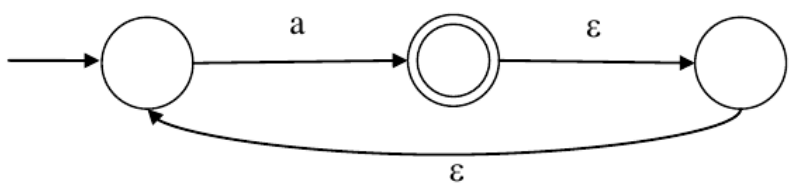
A. The number of a's in w is more than the number of b's in w.
B. The number of a's in w is greater than or equal to the number of b's in w.
C. Every string in w ends with a.
D. w does not contain the substring bb.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 01sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

What is the complement of the language accepted by the NFA shown below?

Assume $\Sigma = \{a\}$ and ϵ is the empty string.



A. ϕ
B. $\{\epsilon\}$
C. a^*
D. $\{a, \epsilon\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

In the Given language $\mathrm{L} = \{ab, aa, baaa\}$, _____ number of strings are in L^*

1. baaaba
2. aabaaaa
3. baaabaaaabaa
4. baaabaaa

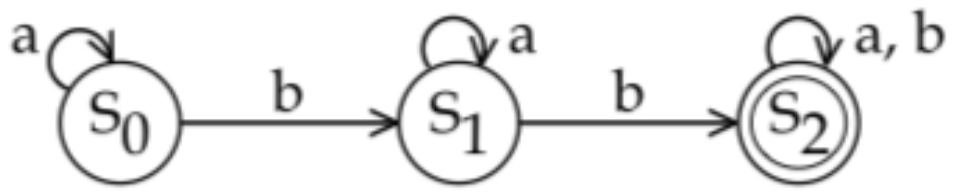
A. 1

B. 2
C. 3
D. 4

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider a DFA machine M whose digraph is :



Then $\mathrm{L}(\mathrm{M})$, the language accepted by the machine M, is the set of all strings having :

A. two or more b's
B. three or more b's
C. two or more a's
D. three or more a's

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

There are exactly _____ different finite automata with three states x, y and z over the alphabet $\{a, b\}$ where x is always the start state.

A. 64
B. 256
C. 1024
D. 5832

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Let $\Sigma = a, b$ and language $\mathrm{L} = aa, bb$. Then, the complement of L is _____.

A. $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* | |w| > 3\}$
B. $\{a, b, ab, ba\} \cup \{w \in \{a, b\}^* | |w| \geq 3\}$
C. $\{w \in \{a, b\}^* | |w| > 3\} \cup \{a, b, ab, ba\}$
D. $\{\lambda, a, b, ab, ba\} \cup \{w \in \{a, b\}^* | |w| \geq 3\}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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